

Swapnil Gautam

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SUMMARY

Machine Learning Engineer with **5+ years of experience building and deploying AI systems**, specializing in **computer vision, deep learning, and Generative AI**. Experienced in developing **end-to-end machine learning pipelines and scalable model deployments** using modern ML frameworks.

EDUCATION

Stevens Institute of Technology (Hoboken, NJ)

Aug 2024 - present

Master of Science in **Applied Artificial Intelligence**

3.8/4

Coursework: Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision

Narsee Monjee Institute of Management Studies

July 2016 - Aug 2020

Bachelor of Technology in Computer Science (Mechatronics)

EXPERIENCE

Conduit (San Francisco, CA)

Machine Learning Engineer

September 2025 - January 2026

- **Designed and deployed** a scalable **computer vision manufacturing process monitoring system** to improve Overall Equipment Effectiveness (OEE), **achieving 86.7% OEE**.
- Built and deployed an end-to-end machine utilization pipeline using YOLO, including model evaluation, benchmarking, **detecting 6+ machine states, and improving production monitoring accuracy by 45%**.
- Fine-tuned and deployed **PaddleOCR on edge devices** for CNC HMI screen text recognition, enabling real-time **data ingestion via MQTT to backend servers**.

WRS Lab (Hoboken, NJ)

AI Research Assistant

January 2025 - August 2025

- Developed a transformer-based model for **Center of Pressure (COP) estimation from an 8-sensor insole + IMU data**, reducing prediction error compared to baseline LSTM models.
- Trained a **transformer model** to estimate the **center of pressure (COP)** from insole sensor, accelerometer, and gyroscope data, and compared its performance with **GPR and LSTM** models for mitochondrial disease assessment.

Wizphys AI (Pune, India)

Founding AI Engineer

April 2022 - June 2024

- **Led the end-to-end development of the ML stack for Wizio**, a real-time physiotherapy app, from concept to final product, including **data preprocessing, model development, testing, POC, and deployment**.
- Designed scalable, robust data pipelines on Firebase (Cloud integration) and a chat interface using state-of-the-art (SOTA) **LLM and RAG (Retrieval-augmented generation) with LangChain**, improved model accuracy, and **reduced hallucinations and processing time by 70%**.
- **Managed and mentored a team of 4 ML engineers**, driving rapid product updates, code review, debugging, and feature enhancements, boosting **patient retention by 140% in 6 months**.

Camweara (Bangalore, India)

Computer Vision Engineer

July 2021 - January 2022

- **Tested and validated** custom-deployed **image segmentation model** using **PyTorch and ONNX**, cutting third-party API dependencies and **reducing costs by 90%**.
- Designed a custom hand-landmark detection model using TensorFlow, **improving landmark localization accuracy and reducing inference latency through quantization**.

SKILLS

- **Programming languages and Frameworks:** Python, C++, PyTorch, Transformers, Scikit-learn, LangChain, RAG, TensorFlow, Hugging Face, AzureML, AWS, Git version control, CI/CD, SQL, MLOps
- **Other Skills:** Strong Problem-solving, Excellent Communication skills (written and verbal), Great collaborative skills, proactive attitude with an eagerness to learn

PROJECTS

Classifying Lumbar Spine Degeneration (Medical Imaging):

- Built **CNN, SVM, and Random Forest models**, achieving **94.4% accuracy** in spinal stenosis classification.
- Developed pipelines to preprocess **35 GB of MRI data**, ensuring consistent and reliable model inputs.
- Used an **ROI-based approach** to crop areas of interest, **improving model accuracy and reliability**.

PUBLICATIONS

- **[Presented conference paper 'Versatile Robotic Arm' at IEEE PuneCon2023](#)**